

Zaid Duraid

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EDUCATION

Memorial University of Newfoundland

St. John's, NL, Canada

Bachelor of Engineering: **Computer Engineering - CGPA: 4.00/4.00 (95%)**

Sep. 2022 - Apr. 2027

- **Dean's List (2022-2025)**
- **Scholarships:** Memorial University Alumni Entrance, Fry Family Foundation Dean's Leadership, Innovasea Computer Engineering, Verafin Inc. Computer Engineering, Charlie Sheppard Memorial - Hatch, PEGNL Past-President, Bob Thorburn Memorial, Gander Lions Club Community Leaders, NLESD Education Foundation

EXPERIENCE

Firmware Validation Engineer Intern

Jan. 2026 - Apr. 2026 and Aug. 2026 - present

Tesla Inc.

Palo Alto, California, USA

- Implemented **LIN in Verilog on a Xilinx FPGA** for Tesla's processor-in-the-loop (PIL) validation, with a Linux driver and fault injection, helping unblock PIL coverage for ~92% of drive inverter (DI) firmware configurations
- Integrated a **MATLAB PMSR motor model onto a Xilinx FPGA in Verilog**, enabling real-time PIL dyno testing for DI firmware configurations not ready for high-load HIL, all within minutes of iteration time
- Developed **fault-handling validation tests** for verifying correct and timely fault responses across DI firmware configurations, including 6+ fault scenarios across gate driver, power, and electrical oil pump systems
- Bringing up a **caliper model in Verilog** for Tesla's PIL validation of the break-by-wire system

Advanced Research Electrical Engineer Intern

May 2025 - Aug. 2025

Solace Power

St. John's, NL, Canada

- Improved the design of a multi-MHz synchronous rectifier by adding a **PLL for closed-loop control** to maintain operation during ~100 us loss of switch gate driver signal
- Designed and built single-layer PCBs to quickly verify **integration of subcircuits** (e.g., RC inverter delay line for phase shift on GaN FET driver signal) into multilayer PCBs
- Researched **EMC solutions for a tens of MHz wireless power system**, reducing noise on key harmonics
- Wrote **C firmware for a TI F28003x microcontroller** to generate internal-clock or interrupt-driven PWM signals up to 9 MHz, with adjustable frequency, duty cycle, and phase shift via CANBUS

FPGA Hardware Engineer Intern

Jan. 2024 - Apr. 2024 and Sep. 2024 - Dec. 2024

Avalon Holographics Inc.

St. John's, NL, Canada

- Led the development of new **internal verification tools using Python** that integrated with existing testbenches to verify hardware image correction features for the holographic display and reduce iteration time by 10x
- Developed and documented **simulation testbenches for FPGA designs in SystemVerilog** that integrate with internal verification tools, reducing critical bug discovery from ~2 months on hardware to ~2-3 days in simulation
- Followed internal test processes by **assembling hardware, executing tests, and running TCL scripts** for the bring-up of new/refurbished PCBs and validation of FPGA bitstream releases

PROJECTS

Part Time Founding Engineer - Typescript, NodeJS, React, React3Fiber, AWS CDK

Dec. 2025 - July 2026

- Contributed to a full-stack cloud application for uploading, hosting, and interacting with 3D GLB scans for occupational therapy home assessments, **supporting patient data for dozens of pilot users**
- Integrated compliant product analytics using PostHog, **providing previously unavailable visibility into daily active users and in-app usage patterns** across development, staging, and production environments

Eastern Edge Robotics Student Design Team Software Co-Lead - Python, C/C++

Aug. 2023 - Dec. 2025

- Built **ROS2 applications on a Raspberry Pi** to interface with ROV hardware over UART and I2C
- Implemented a **Gazebo ROV simulation environment** with C++ plugins for validating the full software stack
- Competed and presented alongside the team at the International MATE ROV Competition Explorer Class in 2023, 2024, and 2025, **placing in 11th, 9th, and 3rd place in the world out of around 20-30 teams**, respectively

SKILLS

Software/Firmware/HDL: Verilog/SystemVerilog, C/C++. Python, Golang, Rust, Typescript

Hardware: Vivado, Quartus, Questa, KiCad, LTSpice, Altium

Tools: Git, Claude Code, Docker, ROS, Gazebo, Solidworks, Onshape